MATH4041/MCSC6120 – Assignment 1: Glucose control, part 1

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1 An ODE with 2 DOF

$$u = \alpha v + \beta \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\,\mathrm{d}t', \qquad \dot v = -\gamma v - \delta u\,(v+\bar v) + F.$$

Differentiate u. By the Leibniz rule the derivative of the integral is $\lambda v(t) - \lambda \int_{-\infty}^{t} \lambda e^{-\lambda(t-t')} v(t')\,\mathrm{d}t'$, and the remaining integral equals $(u-\alpha v)/\beta$ by the first equation. Hence

$$\dot u = \alpha \dot v + \beta\lambda v - \lambda(u - \alpha v) \quad \Longleftrightarrow \quad \dot u + \lambda u = \alpha \dot v + \lambda(\alpha+\beta) v.$$

With $f(u,v) = -\gamma v - \delta u(v+\bar v) + F$ this gives two first-order ODEs

$$\dot u = -\lambda u + \lambda(\alpha+\beta) v + \alpha f(u,v), \qquad \dot v = f(u,v).$$

2 Scaling

Put $v = \bar v\, v'$, $u = \tilde u\, u'$, $t = \tilde t\, t'$, so $\frac{\mathrm{d}}{\mathrm{d}t} = \frac{1}{\tilde t}\frac{\mathrm{d}}{\mathrm{d}t'}$. The v-equation becomes

$$\frac{\mathrm{d}v'}{\mathrm{d}t'} = -(\gamma\tilde t)\, v' - (\delta \tilde t \tilde u)\, u'(v'+1) + \frac{\tilde t}{\bar v} F.$$

$\gamma\tilde t = 1$ and $\delta\tilde t\tilde u = 1$, i.e. $\tilde t = 1/\gamma$ and $\tilde u = \gamma/\delta$, and write $G = F/(\gamma\bar v)$. For the u-equation note that $f = \dot v = \frac{\bar v}{\tilde t}\frac{\mathrm{d}v'}{\mathrm{d}t'} = \frac{\bar v}{\tilde t} f'$. Then

$$\frac{\mathrm{d}u'}{\mathrm{d}t'} = -(\lambda\tilde t)\, u' + \Big[\lambda(\alpha+\beta)\frac{\bar v \tilde t}{\tilde u}\Big] v' + \Big[\alpha\frac{\bar v}{\tilde u}\Big] f', \qquad \text{with} \quad \frac{\tilde t}{\tilde u} = \frac{\delta}{\gamma^2}, \quad \frac{\bar v}{\tilde u} = \frac{\bar v \delta}{\gamma}.$$

$$\boxed{\dot u = -a u + b v + c\, f(u,v), \qquad \dot v = f(u,v) = -v - u(v+1) + G,} \qquad a = \tfrac{\lambda}{\gamma},\ b = \tfrac{\lambda(\alpha+\beta)\bar v\delta}{\gamma^2},\ c = \tfrac{\alpha\bar v\delta}{\gamma},\ G = \tfrac{F}{\gamma\bar v}.$$

Seven parameters are reduced to four, all dimensionless (a unit check confirms this; it also shows that the factors $\tilde t/\tilde u$ and $\bar v/\tilde u$ must not be dropped). In these variables $v = 0$ is the baseline and $v = -1$ means zero glucose, so only $v > -1$ is physical.

3 Equilibria

$\dot u = \dot v = 0$ means $f = 0$ and $-au + bv = 0$, so $u^* = \frac{b}{a} v^*$ and $-v^* - \frac{b}{a} v^*(v^*+1) + G = 0$, i.e. $v^{*2} + (1 + \frac{a}{b}) v^* - \frac{a}{b} G = 0$. Hence

$$v^*_{\pm} = -\frac{1}{2}\Big(1+\frac{a}{b}\Big) \pm \frac{1}{2}\sqrt{\Big(1+\frac{a}{b}\Big)^2 + \frac{4aG}{b}} = \frac{-(a+b) \pm \sqrt{D}}{2b}, \qquad u^*_{\pm} = \frac{b}{a} v^*_{\pm}, \qquad D = (a+b)^2 + 4abG.$$

There are two equilibria if $G > G_{SN}$, one if $G = G_{SN}$ and none if $G < G_{SN}$, where (saddle-node point)

$$G_{SN} = -\Big(1+\frac{a}{b}\Big)^2 \frac{b}{4a} = -\frac{(a+b)^2}{4ab}, \qquad v^* = -\frac{a+b}{2b} \ \text{ at } G = G_{SN}.$$

$G = v^* + \frac{b}{a} v^*(v^*+1)$, a parabola in v^*. So the branch is a parabola that opens to the right, with its tip at G_{SN} (Fig. 1a). It passes through $v^* = 0$ and $v^* = -1 - \frac{a}{b}$ at $G = 0$, and through $v^* = -\frac{a}{b}$ and $v^* = -1$ at $G = -1$. The parameter c does not change the equilibria. For $a = 0.1$, $b = 1$: $G_{SN} = -3.025$.

4 For what parameters and initial conditions is the model controllable?

with F (so G) constant, we use three levels: **C1** $v(t)$ stays bounded; **C2** $-1 < v(t) < 3$ for all t (glucose stays positive and below $4\bar v$); **C3** C2 and $v(t)$ does not oscillate.

4.1 Stability of the equilibria

The Jacobian is $J = \begin{pmatrix} -a - c(v+1) & b - c(1+u) \\ -(v+1) & -(1+u) \end{pmatrix}$ and the eigenvalues solve $\lambda^2 - \operatorname{tr} J\,\lambda + \det J = 0$. At an equilibrium we can use $au^* = bv^*$:

$$\det J = a(1+u^*) + b(1+v^*) = a + b + 2bv^* = \pm\sqrt{D}, \qquad \operatorname{tr} J = -(1+a+c) - \Big(c + \frac{b}{a}\Big) v^*.$$

- *Lower equilibrium*: $\det J = -\sqrt{D} < 0$, so it is always a saddle.
- *Upper equilibrium*: $\det J = \sqrt{D} > 0$, so it is stable if and only if $\operatorname{tr} J < 0$, i.e. $v^*_+ > v_H = -\frac{a(1+a+c)}{ac+b}$, which means $G > G_H = v_H + \frac{b}{a} v_H(v_H+1)$. For $a = 0.1$, $b = c = 1$: $G_H = -1.736$.
- *Zero-glucose line*: on $v = -1$ we have $\dot v = 1 + G$ for every u. For $G > -1$ orbits can only cross it upward, so glucose never reaches zero. For $G < -1$ they can only cross downward, so once glucose reaches zero it never recovers.

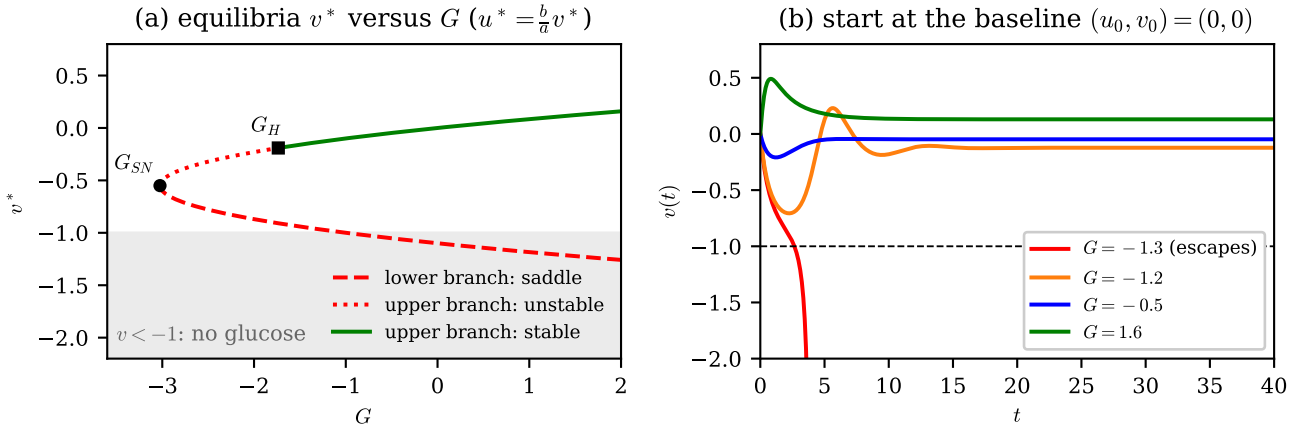


Figure 1: $a = 0.1$, $b = c = 1$. (a) The branch of equilibria; the grey band is negative glucose. (b) $v(t)$ for a patient who starts at the baseline $(0, 0)$.

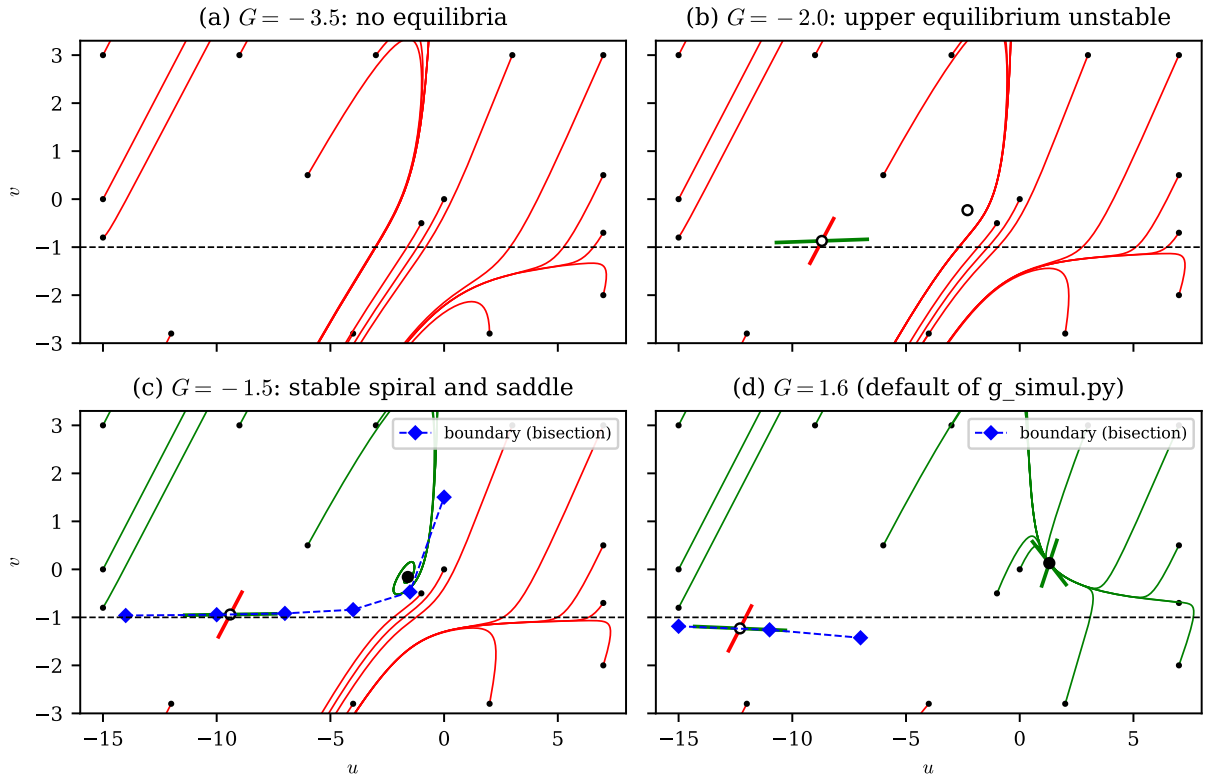


Figure 2: Phase portraits ($a = 0.1$, $b = c = 1$). Black dots: initial conditions; green orbits stay bounded, red orbits escape. Circles: equilibria (filled = stable) with eigenvector lines as in `g_simul.py`. Blue: boundary of the controlled region, found by bisection in v_0.

4.2 Experiments with `g_simul.py`

All runs use `RK2step` from `g_simul.py` with $h = 0.001$ (halving h changes the orbits by less than 10^{-5}) and the same stopping rule: an orbit *escapes* if $v < -100$. Default parameters: $a = 0.1$, $b = c = 1$.

Table 1: Equilibria and eigenvalues, as printed by `g_simul.py` ($a = 0.1$, $b = c = 1$). For $G = -3.5$, $D < 0$: no equilibria.

G	upper (u^*, v^*)	eigenvalues	type	lower (u^*, v^*)	eigenvalues (saddle)
-2.5	$(-3.209, -0.321)$	$0.485,\ 0.944$	unstable node	$(-7.791, -0.779)$	$-0.070,\ 6.540$
-2.0	$(-2.298, -0.230)$	$0.214 \pm 0.771i$	unstable spiral	$(-8.702, -0.870)$	$-0.085,\ 7.556$
-1.5	$(-1.595, -0.159)$	$-0.173 \pm 0.867i$	stable spiral	$(-9.405, -0.941)$	$-0.094,\ 8.339$
-0.5	$(-0.475, -0.048)$	$-0.789 \pm 0.619i$	stable spiral	$(-10.525, -1.052)$	$-0.105,\ 9.582$
1.6	$(1.301,\ 0.130)$	$-0.440,\ -3.091$	stable node	$(-12.301, -1.230)$	$-0.118,\ 11.549$

Phase portraits (Fig. 2). (a) $G < G_{SN}$: there are no equilibria and every orbit escapes, (b) $G_{SN} < G < G_H$: both equilibria are unstable and every orbit we tried escapes. (c) $G_H < G < -1$: a stable spiral and a saddle with $v^* > -1$. Orbits on the upper-left side of a boundary curve through the saddle go to the spiral; the others escape. We located this boundary by bisection in v_0 for fixed u_0, the way the comment in `g_simul.py` suggests; for $G = 2$,

$u_0 = -11$ this gives $v_0 = -1.29943$, the value in that comment. (d) $G > -1$: the saddle and the boundary lie in $v < -1$, and every physical initial condition we tried is controlled.

Table 2: Experiments from the baseline $(u_0, v_0) = (0, 0)$, $a = 0.1$, $b = c = 1$. "Extrema" counts the maxima and minima of $v(t)$; we count C3 as satisfied if there is at most one.

G	outcome	min v	max v	extrema	C1	C2	C3
$-3.5, -2.0, -1.5, -1.3$	escapes	$-\infty$	0	–	no	no	no
-1.2	bounded	-0.707	0.230	10	yes	yes	no
-1.0	bounded	-0.500	0	7	yes	yes	no
-0.5	bounded	-0.209	0	3	yes	yes	no
0	at equilibrium	0	0	0	yes	yes	yes
1.6	bounded	0	0.491	1	yes	yes	yes
5.0	bounded	0	1.217	1	yes	yes	yes
20.0	bounded	0	3.181	1	yes	no	no

Patient at baseline (Table 2, Fig. 1b). For $G \leq -1.3$ glucose drops through zero and escapes. For $-1.2 \leq G < 0$ the patient is controlled but v oscillates while it settles (the upper equilibrium is a stable spiral). For moderate $G > 0$ it settles without oscillating (stable node). For very large input ($G = 20$) glucose overshoots $v = 3$ before settling. By bisection in G, the baseline stays bounded exactly for $G > G' = -1.254$.

Other parameters (Table 3). A larger instantaneous gain c does not move the equilibria, but it lowers G_H and G': the patient can tolerate more exertion. Changing a moves G_{SN} a lot but G' only a little. In all cases $G_H < G' < -1$.

Table 3: Thresholds ($b = 1$). G': smallest G for which the baseline stays bounded (bisection).

a	c	G_{SN}	G_H	G'
0.1	0	-3.025	-1.089	-1.062
0.1	1	-3.025	-1.736	-1.254
0.1	3	-3.025	-2.475	-1.618
0.3	1	-1.408	-1.361	-1.182

4.3 Conclusions

1. For $G < G_{SN} = -(a+b)^2/(4ab)$ the model is not controllable: there are no equilibria and every orbit escapes.
2. For $G_{SN} < G < G_H$ the model is not controllable: both equilibria are unstable and all orbits we tried escape ($G_H = -1.736$ for the default parameters).
3. For $G_H < G < -1$ the model is controllable only for initial conditions on the upper-left side of the boundary through the saddle. A patient starting at the baseline is controlled if and only if $G > G'$ ($G' = -1.254$ for the default parameters).
4. For $G > -1$ every initial condition with positive glucose ($v_0 > -1$) is controlled (C1) and glucose never reaches zero. The approach oscillates for G below about 0 and is monotone for moderate $G > 0$ (C3); C2 fails only for very large G (glucose spike) or a very negative initial control u_0.
5. A stronger instantaneous response c makes the model controllable for a larger range of G.